第43屆國際大學程式設計競賽－亞洲區台北站
The 43rd Annual International Collegiate Programming Contest
- Asia Taipei Regional Contest

Asia Taipei Regional Contest

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- **Problems**: There are 11 problems in this contest. (41 pages including the cover in total).
- **Input**: Input of all the problems are from the standard input (`stdin`). Multiple test cases may be given according to the instructions of each problem.
- **Output**: All outputs should be directed to the standard output (`stdout`).
- **Time Limits**: Contest judges will execute the submitted programs within certain time limits (given in the table below).
- **Memory Limits**: The submitted programs will be executed in a specific sandbox. The sandbox will allocate 256 MB of memory. Please note that if your problem exceeds the memory limit, you may receive any verdicts other than `Yes` (including `Wrong-Answer`, `Runtime-Error`).

Problem Information

Problem	Problem Name	Runtime Limit
A	Automatic Control Machine	1 second
B	No Tabs	2 seconds
C	Starlight on the Stage	4 seconds
D	Dictionary Problem	2 seconds
E	Pairing Points in Three-Dimensional Spaces	5 seconds
F	Will You Save Us?	1 second
G	Minimum Shipping Cost	2 seconds
H	Hidden Glyphs	8 seconds
I	Protein Structures	5 seconds
J	Mean Weight of a Critical Cycle	3 seconds
K	Monochrome Land	1 second

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Problem A.
Automatic Control Machine

Time limit: 1 second

Angela is very envious of beautiful long hair. She is very annoyed that her hair grows too slow. Through a friend's suggestion, she found a magical shampoo that claims to make hair grow fast and long. Angela found that each time she used the magical shampoo, her hair would be 10% longer the next day. Angela regularly shampoos her hair once every 3 days, and she uses the magical shampoo whenever she washes her hair. However, Angela's family loves to travel. Every two weeks, they will use the weekend to travel for two days. If she encounters regular shampoo on those two days, Angela can only use the regular shampoo given by the hotel. Angela just came back from a family tour yesterday. Let today be the first day, as the family travel tour is scheduled on every other weekend, next trip will be on the 13th and 14th days. Suppose that her current hair length is 20 cm and she starts to use the magical shampoo starting from today. On the second day, her hair length will become 22 cm. On the fourth day she will wash her hair again and her hair has a length of 24.2 cm on the fifth day, etc. Angela's hair length is illustrated as Table 1.

Table 1: The day-by-day variation of Angela's hair length.

Day	Hair length	Action		Day	Hair length	Action
1	20	shampoo		13	29.282	shampoo, travel
2	22			14	29.282	travel
3	22			15	29.282	
4	22	shampoo		16	29.282	shampoo
5	24.2			17	32.2102	
6	24.2			18	32.2102	
7	24.2	shampoo		19	32.2102	shampoo
8	26.62			20	35.43122	
9	26.62			21	35.43122	
10	26.62	shampoo		22	35.43122	shampoo
11	29.282			23	38.974342	
12	29.282			24	38.974342	

By Table 1, Angela's hair will be more than 30 cm on the 17th day and more than 35 cm from the 20th day. Please help Angela to find on which day her hair will reach (or beyond) a certain length.

Input

The first line contains an integer T indicating the number of test cases, following T lines each contains two space-separated integers A and B indicating the length of Angela's hair now and the desired length (both in cm), respectively.

- $1 \leq T \leq 10$
- $10 \leq A \leq B \leq 2147483647$

Output

For each test case, please output one line containing only one integer indicating the day number of the first day for Angela to have hair length reach (or beyond) the desired length.

Sample Input	Sample Output
1 20 30	17

Sample Input	Sample Output
2 20 35 10 20	20 26

Problem B.
No Tabs

Time limit: 2 seconds

Disaster is coming! A virus named **TABX-4869** is developed by a mad scientist. **TABX-4869** can infect all creatures on a planet instantly. People infected with **TABX-4869** will always enter *TAB* when they want to enter *SPACE*!

The mad scientist plans to spread **TABX-4869** to the whole galaxy through star routes. The galaxy contains N planets and M undirected routes. **TABX-4869** can spread to all neighbor planets at a time, but it needs D time unit(s) when spread through a route with distance D. You can assume that the galaxy is connected via star routes.

To prevent the worst, the Galaxy Government (**GG**) developed the vaccine and want to spread it as soon as possible. The vaccine can prevent uninfected creature from getting infected but cannot cure infected creatures. In addition, the vaccine can be spread rapidly exactly like **TABX-4869**.

GG has observed that the mad scientist will begin to spread **TABX-4869** from a particular planet as source. Unfortunately, **GG** cannot predict which planet is the source. Due to the limited amount of vaccine, **GG** can only select exactly one planet as the source of vaccine as soon as the source of **TABX-4869** is observed.. (You can assume that **GG** and the mad scientist will start spreading both at the same time.)

There are many great engineers and mathematicians in **GG**, so **GG** will always find the best vaccine source to minimize the disaster. However, they may not save all planets if the mad scientist is smart enough. Please help **GG** to compute the largest possible number of planets that will be infected.

Note that a planet will be infected even if **TABX-4869** and the vaccine are spread to it at the same time.

Input

The first line contains an integer T indicating the number of test cases, described by the following lines.

For each test case, the first line contains two space-separated integers N and M. Each of the following M lines contains three space-separated integers A, B, D that denote a route between planet A and planet B with distance D. Planets are numbered from 1 to N. Each pair of planets has at most one star route between them.

- $1 \leq T \leq 20$
- $3 \leq N \leq 200$
- $N \leq M \leq \frac{N(N-1)}{2}$
- $0 < D < 10000$

Output

For each test case, please output one line containing only one integer indicating the largest number of planet will be infected.

Sample Input	Sample Output
2 3 3 1 2 5 2 3 8 3 1 6 5 5 1 2 1 1 3 1 1 4 1 1 5 1 3 4 1	2 4

Sample Explanations

- For the first sample, at least 2 planets will be infected if the **TABX-4869** source is planet 1. (If the **TABX-4869** source is not planet 1, we can save more planets. Hooray!)
- For the second sample, at least 4 planets will be infected if the **TABX-4869** source is planet 1.

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Problem C.
Starlight on the Stage

Time limit: 4 seconds

Recently, talent shows become very popular worldwide. This year, in a small isle country on the Pacific Ocean, a talent contest called '*Internatinoal Classic Performance Competition* (*ICPC*)' is held, with the main topic 'musical drama'. During the contest, the participants will be paired to perform *duets* (i.e., duo dances) against their partners on the stage. The top two winners recieve the prize, including an invitation to participate real musical concerts as an actor/actress.

Flora and Claire are best friends of each other since childhood. They are currently studying in an academy for performing arts. After some discussions, Flora and Claire decided to participate the competition and aim at becoming the top stars among all the participants.

The competition is formed in a two-phase schedule, and there are N participants in total. (Flora is the first participant; Claire is the N-th participant.) The first phase has ended several days ago, where the judges have annouced the initial *scores* evaluated based on the proposed resumes. The score of the i-th participant is denoted as s_i, and its initial value is available from the official announcement. Then, the second phase will be processed with following rules:

- There are N rounds that will be held.
- In the i-th round, the i-th participant will be paired with every other participant exactly once. That is, $(N-1)$ duet stages will be performed in each round.
- For each duet stage, to make the problem easier, we assume that the winner will always be the participant who has the higher score at that time point. That is, for the duet stage involving the x-th participant and the y-th participant, if $s_x > s_y$, then the x-th participant always win against the y-th participant. If the scores are same, the stage will be tied (i.e., draw game).
- After the i-th round, suppose that the result of the i-th participant is W wins, L losses and D tied games (draws), then the score of the i-th

participant will be updated as $s_i^{\text{new}} := s_i + W - L$. (Note that only s_i will be updated for the i-th round.)

- The N rounds will be held strictly in order, from $i = 1, 2, \ldots$ to N.
- After all the N rounds, the winners will be the two participants having highest final scores.

(Note that, the score of each participant will be changed *exactly once* after the procedure.)

Before the second phase of competition, Flora and Claire decide to conduct additional dancing training so that they can result in top two scores after all the N rounds of stages. However, Flora also do not want to waste her stamina too much since the second phase competition is nearing the date (while it is not an issue for Claire).

Assume that after the training, the initial score of Flora (i.e., initial s_1) will be increased by α, and the initial score of Claire (i.e., initial s_N) will be increased by β. Please help Flora and Claire to achieve the goal with minimum α, where α and β must be non-negative integers.

Moreover, we only consider the cases that Flora and Claire result in top scores *strictly higher than all the other participants* as valid cases.

Input

The first line of input contains an integer T indicating the number of test cases. The following lines describe T cases in order.

For each test case, the first line will contains an integer N, indicating the number of participants. Then, each of the following N lines will contain an integer denoting the initial value of s_i based on the official announcement, for $i = 1, \ldots, N$, respectively.

- $1 \leq T \leq 10$
- $3 \leq N \leq 10000$
- All the initial values of s_i are positive integers no larger than 10^8, $\forall i$.

Output

For each test case, please output one line containing an integer indicating the minimum valid α.

Sample Input	Sample Output
1 3 100 101 105	2

Sample Input	Sample Output
2 5 99999 2018 11 10 99999 3 10000007 10000009 10000005	0 3

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Problem D.
Dictionary Problem

Time limit: 2 seconds

Imagine a world equipped with character set $\mathbf{U} = \{c_1, c_2, \ldots, c_k\}$ with an unknown total order $\prec_{\mathbf{U}}$. You are given an important string $S = \langle s_1, s_2, \ldots, s_n \rangle$ consists of characters from set $\mathbf{U}$. Also, the lexicographical order of **suffices** of S is given. Please write a program to compute a possible total order $\prec'_{\mathbf{U}}$ (maybe not identical to the unknown $\prec_{\mathbf{U}}$) of $\mathbf{U}$.

Let $S[i, j]$ with $1 \leq i \leq j \leq |S|$ be the string $\langle s_i, s_{i+1}, \ldots, s_j \rangle$, where $|S|$ is the length of S. The **suffices** of S is a particular set of strings $\mathbf{suf}(S) := \{S[i, |S|] \mid 1 \leq i \leq |S|\}$. For example, the suffices of string `"abb"` are `"abb"`, `"bb"` and `"b"`.

We call that a string X is lexicographically smaller than a string Y (written as $X \prec^*_{\mathbf{U}} Y$) if either X is exactly $Y[1, |X|]$ or $X_i \prec_{\mathbf{U}} Y_i$, where i is the first position having different characters in X and Y. For example, consider the character set of lowercase letters with the common character order from `a` to `z`, then `"aa"` is lexicographically smaller than `"aab"`; `"acc"` is lexicographically smaller than `"baa"`.

For this problem, you will be given the rank of each suffix of S. The smaller the rank is, the smaller the lexicographical order is. Suppose that we are given the order that `b` is smaller than `a`. String `"abba"` has its smallest suffix being `"bba"` and the biggest one being `"abba"`. To construct the rank one may have multiple choices for order of $\mathbf{U}$. Both `b < c < a` or `c < b < a` act as a valid choice.

Table 2: Rank of the suffices of `"abba"`.

Suffix	Rank
`"abba"`	4
`"bba"`	1
`"ba"`	2
`"a"`	3

Input

The first line of input contains an integer T indicating the number of test cases, described by the following lines.

For each test case, there are three lines of input. The first line of the test case contains two space-separated integers n and k. The second line of the test case has n space-separated integers $a_1, a_2, \ldots, a_n$ representing the important string $S = \langle c_{a_1} c_{a_2} \ldots c_{a_n} \rangle$. The third line of the test case has n space-separated distinct integers $r_1, r_2, \ldots, r_n$ representing the rank of each suffix of S, where r_i is the rank of $S[i, n]$ among $\mathbf{suf}(S)$.

- $1 \leq T \leq 100$
- $2 \leq n, k \leq 200000$
- $1 \leq a_i \leq k$ for every $i = 1, 2, \ldots, n$
- $1 \leq r_i \leq n$

Output

For each test case, please output one line containing k integers $b_1, b_2, \ldots, b_k$, separated by single spaces, to indicate c_{b_1} being the smallest character of $\mathbf{U}$, c_{b_2} the second smallest and so on.

If there are multiple solutions, output any of them. We guarantee that there must be a valid solution for every test case.

Sample Input	Sample Output
2 4 2 1 2 2 1 4 1 2 3 3 3 1 1 1 3 2 1	2 1 3 2 1

Sample Input	Sample Output
2 3 3 1 2 3 1 2 3 3 3 2 1 3 1 2 3	1 2 3 2 1 3

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Problem E.
Pairing Points in Three-Dimensional Spaces

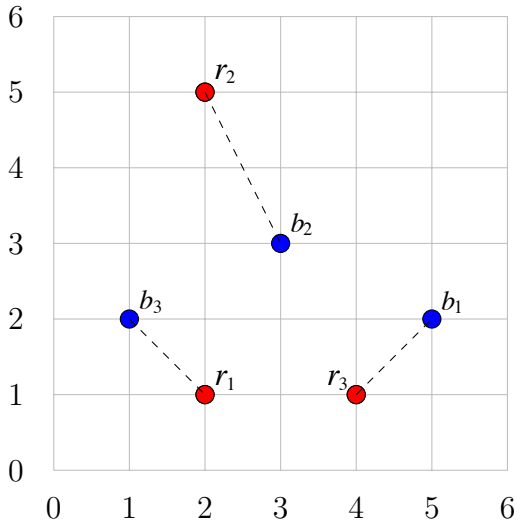
Time limit: 5 seconds

We are particularly concerned with the problem of pairing points in three-dimensional spaces. There are two sets of equally sized points, each with coordinates (x, y, z), and we need to group these points into pairs. The cost of pairing two points is the *squared Euclidean distance* between them. The goal is to find the optimal pairing such that the total cost is minimized.

Formally, there are two sets of n points; namely red points $R = \{r_i, r_2, \ldots, r_n\}$ and blue points $B = \{b_1, b_2, \ldots, b_n\}$, where $r_i, b_i \in \mathbb{R}^3$ for all $1 \le i \le n$. The cost function between two points p and q is defined by

$$\text{cost}(p, q) = (p_x - q_x)^2 + (p_y - q_y)^2 + (p_z - q_z)^2.$$

For example, consider the the following two sets of 3 points; here the red points $R = \{r_1 = (2, 1, 0), r_2 = (2, 5, 0), r_3 = (4, 1, 0)\}$ and blue points $B = \{b_1 = (5, 2, 0), b_2 = (3, 3, 0), b_3 = (1, 2, 0)\}$.



(An illustration of the plane $z = 0$.)

It can be easily verified that the corresponding minimum cost can be achieved by pairing b_1 with r_3, b_2 with r_2, and b_3 with r_1; note that the minimum total costs in the pairing is $2 + 2 + 5 = 9$.

Input

The first line of input contains an integer T indicating the number of test cases. The following lines describe T cases in order.

For each test case, the first line contains an integer n indicating the number of points in sets R and B. Subsequently, it is followed by $n+n=2n$ lines. Each of the first n lines contains three integers representing the (x, y, z)-coordinates of the red points; each of the other n lines contains three integers representing the coordinates of the blue points.

Note that the given points may have same coordinates of (x, y, z).

- $1 \leq T \leq 20$
- $1 \leq n \leq 250$
- $0 \leq x, y, z \leq 40000$ for all the given points.

Output

For the output for each test case, please print out an non-negative integer as the the minimum total costs in the optimum pairing.

Sample Input	Sample Output
1 3 2 1 0 2 5 0 4 1 0 5 2 0 3 3 0 1 2 0	9

Sample Input	Sample Output
2 1 2018 2018 2018 2019 2019 2019 2 0 0 100 0 0 100 0 0 100 0 0 100	3 0

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Problem F.
Will You Save Us?

Time limit: 1 second

You, the hero, are going to fight with the final boss of the world, Irisu. As a hero, you have three character "abilities": Body B, Mind M, and Intelligence I, where the overall strength S is calculated as the geometric mean of these abilities. In other words, $S = \sqrt[3]{BMI}$. Unfortunately, because you failed the last battle against the boss, all your abilities are zero now.

To beat Irisu, your overall strength should satisfy a mysterious inequality $(S+5)(S+1)(S+4) \geq X$ on the elder scroll you found in a huge tomato after the last battle. The only way you can increase the abilities is to buy ability potions from the dark alchemist, Skyly. There are $N_B + N_M + N_I$ potions which Skyly is selling. The first N_B ones are Body Potions, the following N_M ones are Mind Potions, and the last N_I ones are Intelligence Potions. The i-th potion would cost you P_i dollars and increase the corresponding ability by E_i points.

To save both the world and the wallet, you want to find the minimum total price you need to spend for the ability potions to beat the final boss.

Input

The first line contains four integers X, N_B, N_M, N_I, and the i-th line of the following $N_B + N_M + N_I$ lines contains two integers P_i, E_i.

- $0 \leq X \leq 10^9$
- $0 \leq N_B, N_M, N_I \leq 30$
- $0 \leq P_i, E_i \leq 10^9$ for all $1 \leq i \leq N_B + N_M + N_I$

Output

Please output the minimum total price you need to spend for the ability potions to beat the final boss. If you are still not strong enough even you bought all the potions from Skyly, please output "`DEAREST DROP`".

Sample Input	Sample Output
31 4 1 5 9 2 6 5 3 5 8 9 7 9 3 2 3 8 4 6 2 6 4 3	12

Sample Input	Sample Output
5566 2 1 0 0 999999999 999999999 0 0 0	DEAREST DROP

Sample Input	Sample Output
50216 2 2 2 1 20 1 30 10 42 20 43 15 18 17 19	29

Problem G.
Minimum Shipping Cost

Time limit: 2 seconds

Pigou works at Inter-Continental Prado Cargo (ICPC) as a manager to arrange the shipping business of cargos. To model the problem, Pigou begins with a simple example as in Figure 1. There are two arcs a_1 and a_2 indicating there are two methods to ship cargo from the origin O to the destination D, however, these two ways of shipping might lead to different costs.

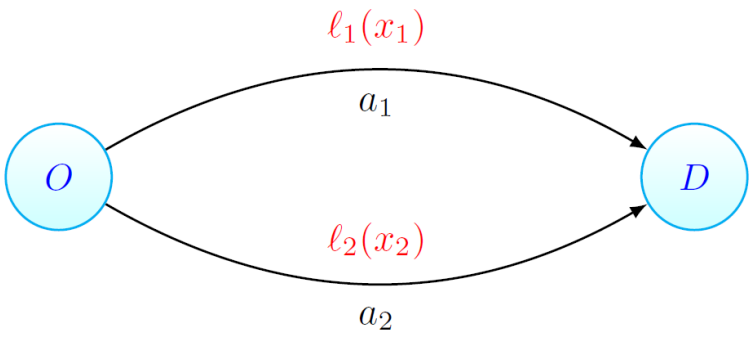


Figure 1: A simple example.

Define the cost of the shipping methods by cost functions: if a fraction x_1 of the cargos use arc a_1, then $\ell_1(x_1)$ represents the cost of the cargo to reach the destination; similarly, if a fraction x_2 of the cargos use the alternative arc a_2, then $\ell_2(x_2)$ represents the incurred cost. Naturally, the higher the fraction, the higher the cost. So assume that $\ell_1(x_1)$ and $\ell_2(x_2)$ are non-decreasing with respect to x_1 and x_2, respectively.

An efficient use of the methods means to minimize the total cost, which in this case can be expressed as the minimization of $x_1\ell_1(x_1) + x_2\ell_2(x_2)$, where x_1 is the fraction of cargos that use a_1, and x_2 is the fraction of cargos that use a_2; as they are fractions, we may impose that $x_1 + x_2 = 1$ and $x_1, x_2 \geq 0$. For example, take $\ell_1(x_1) = x_1$ and $\ell_2(x_2) = 1$. In this way, we get $x_1 = 0.5$ and $x_2 = 0.5$, and 0.75 is the minimum cost for this example. This is the so called *optimum*.

Indeed, Pigou considers a more general situation, where there are N different ways (arcs) for shipping: $a_1, \ldots, a_N$ and for $i = 1..N$, a_i has $\ell_i(x)$ as its cost function. Suppose the total cargo to be shipped is 1 and x_i indicates the fraction assigned to arc a_i, for $i = 1..N$. Note that $x_1 + \cdots + x_N = 1$. Pigou further restricts each $\ell_i(x)$ to the form $a_i x^{e_i}$, where a_i is a positive integer and e_i is a non-negative integer. He wants to know how to arrange the x_i's such that $\sum_{i=1}^{N} x_i \ell_i(x_i)$ is minimum.

Given the cost functions, your task is to write a program to help Pigou find the minimum cost.

Input

The first line contains an integer T indicating the number of test cases.

Each case starts with a line of one positive integers, N, which indicate the number of arcs, respectively. Then N lines follow and each consists of two integers, $a > 0$ and $e \geq 0$ indicating that the corresponding cost function $\ell(x) = ax^e$.

- $1 \leq T \leq 10$
- $1 \leq N \leq 100$
- $1 \leq a_i \leq 1000; 0 \leq e_i \leq 5$, for $i = 1, \ldots, N$

Output

For each test case, output one line containing the minimum value of $\sum_{i=1}^{N} x_i \ell(x_i)$. The precision of the answer is 6 places after decimal point without rounding. The error can be at most 10^{-6}.

Sample Input	Sample Output
2 2 1 1 1 0 3 1 2 1 0 2 0	0.750000 0.615100

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Problem H.
Hidden Glyphs
Time limit: 8 seconds

Glyph Yuji loves traveling and discovering various glyphs in different countries. After several years of travelling, Glyph find that there are similarities between glyphs he found in two countries: Lollipop and Boomerang. The handwritting system from these two places are so similar – all glyphs used for handwritting are in rectangular shapes.

A rectangular shaped glyph can be modeled as a simple polygon in 2D Euclidean plane where every edge of the polygon is either parallel or perpendicular to the axes.

With deeper inspection between two languages used in Lollipop and Boomerang, Glyph has made some observations: (1) glyphs from Lollipop are usually larger than glyphs from Boomerang. (2) There are exactly n different glyphs from each language, and there is a one-to-one correspondence between them.

However, Glyph does not know which glyph in Lollipop language corresponds to which glyph in Boomerang language. To explore this correspon-

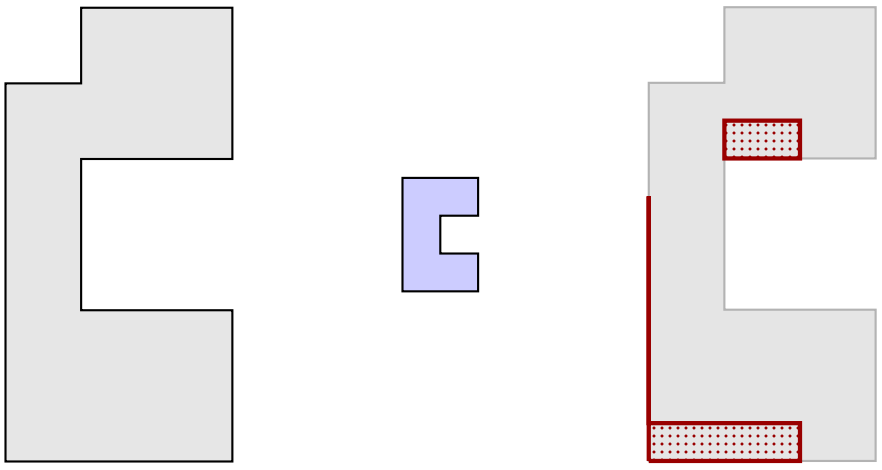


Figure 1: The glyph from Lollipop country (left), the corresponding glyph from Boomerang country (middle) and the possible translation area (right). (With the left-bottom corner as $(0, 0)$)

dence, Glyph first describe each glyph as a simple polygon.

Formally, for a region B and a point (x, y), Glyph define $B + (x, y)$ be the set $B + (x, y) := \{(b_x + x, b_y + y) \mid (b_x, b_y) \in B\}$. For every pair of glyph L in Lollipop and glyph B in Boomerang, Glyph define a "translation area" $tr(L, B)$ to be the area of the region $tr(L, B) := \{(x, y) \mid B + (x, y) \subseteq L\}$. See Figure 1 for example.

Glyph Yuji believes that the true correspondence between the two languages must give the largest total sum of translation area among all matched glyph pairs. Notice that, under this definition it is possible for a pair of glyphs having translation area 0. Can you help Glyph to find this value?

Input

The first line contains an integer T indicating the number of test cases.

Each test case begins with an integer n. Then $2n$ lines follows. The first n lines describe n rectangular shapes representing glyphs from Lollipop language, whereas the following n lines describe n rectangular shapes representing glyphs from Boomerang language.

For the i-th line that describes a rectangular shape, there is an integer m_i denoting the number of vertices followed by m_i pair of integer coordinates listed in either clockwise or counter-clockwise order.

- $1 \leq T \leq 4$
- $1 \leq n \leq 30$
- $4 \leq m_i \leq 200$
- For all coordinates (x, y), $0 \leq x, y \leq 1000$.
- All rectangular shapes are valid, non-degenerate simple polygons.

Output

For each testcase, please output one line containing only one integer indicating the largest total sum of all translation area among matched pairs. It can be shown that the solution must be an integer.

Sample Input	Sample Output
2 1 10 0 0 0 10 2 10 2 12 6 12 6 8 2 8 2 4 6 4 6 0 8 0 0 0 3 2 3 2 2 1 2 1 1 2 1 2 0 2 5 0 0 0 2 0 5 5 5 5 0 4 0 0 3 0 3 6 0 6 4 0 0 1 0 1 2 0 2 4 0 0 0 1 2 1 2 0	6 20

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Problem I.
Protein Structures

Time limit: 5 seconds

A protein is a compound made of amino acids, which are linked into a chain. The determination of the amino acid sequence of a protein is an important step toward quantifying this protein and solving its structure and function. However, such a determination is difficult.

Recently, one scientist found that the amino acid sequence of a given protein can be determined by firstly performing a number of experiments on an (expensive) equipment to obtain some useful information and then solving an optimization problem as follows.

The input of the problem consists of a positive integer t and two sets of integers $A = \{a_1, a_2, ..., a_m\}$ and $X = \{x_1, x_2, ..., x_{n-1}\}$, where $0 < a_i < t$ for each $a_i \in A$ and $0 < x_i < t/2$ for $x_i \in X$. Image that we want to place $n-1$ points $p_1, p_2, ..., p_{n-1}$ on a number line. A placement is *valid* if the position of each p_i is either x_i or $t - x_i$.

For example, let $t = 10$, $n = 4$, and $X = \{2, 3, 4\}$. In this example, p_1 can be placed at either 2 or 8; p_2 can be placed at either 3 or 7; and p_3 can be placed at either 4 or 6. Figure 1 shows two valid placements for this example.

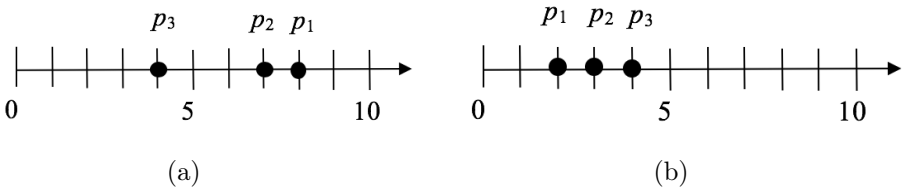


Figure 1: Two valid placements for $t = 10$ and $X = \{2, 3, 4\}$.

Consider a valid placement P. Let $(s_1, s_2, ..., s_{n-1})$ be the sorted sequence of the positions of the $n-1$ points and let $D(P) = (d_1, d_2, ..., d_n)$ be the sequence in which $d_k = s_k - s_{k-1}$ for $1 \le k \le n$, where $s_0 = 0$ and $s_n = t$. For example, in Figure 1(a), we have $(s_1, s_2, s_3) = (4, 7, 8)$ and $(d_1, d_2, d_3, d_4) = (4,$

3, 1, 2); and in Figure 1(b), we have $(s_1, s_2, s_3) = (2, 3, 4)$ and $(d_1, d_2, d_3, d_4) = (2, 1, 1, 6)$.

Different amino acids have different weights. In our application, A represents the set of weights of various known amino acids, n is the number of amino acids in the given protein, and t is the weight of the protein. The set X is produced at the experimental stage. Ideally, there are no errors at the experimental stage and there exists a valid placement P such that the k-th number in $D(P)$, d_k, is just the weight of the k-th amino acid of P; and our problem is to find a valid placement such that all numbers in $D(P)$ are weights in A.

However, in practice, since there may be errors at the experimental stage, our problem is to find a valid placement P such that for each d_k in $D(P)$ there is a number in A that is close to d_k. More formally, define the error of each d_k, $1 \le k \le n$, to be

$$e(d_k) = \min_{1 \le i \le m} |d_k - a_i|$$

and define the error of P to be

$$E(P) = \sum_{1 \le k \le n} e(d_k)$$

For example, let $m = 3$ and $A = \{1, 4, 7\}$. Then, in Figure 1(a), we have $e(d_1) = e(4) = |4 - 4| = 0$, $e(d_2) = e(3) = |3 - 4| = 1$, $e(d_3) = e(1) = |1 - 1| = 0$, and $e(d_4) = e(2) = |2 - 1| = 1$. Thus, for the placement in Figure 1(a), we have $E(P) = 2$. Similarly, it can be seen that for the placement in Figure 1(b), we have $E(P) = 1 + 0 + 0 + 1 = 2$. The number of valid placements is 2^{n-1}. Our problem is to find a valid placement P that minimizes $E(P)$.

Input

The first line contains an integer T indicating the number of test cases.

Each test case begins with a line containing the three integers t, m, and n. Then, two lines follow, where the first line gives the m integers of A and the second line gives the $n - 1$ integers of X.

- $1 \leq T \leq 10$
- $2 \leq t \leq 10^7$
- $2 \leq m \leq 10^6$
- $2 \leq n \leq 3000$
- All numbers in A are (distinct) integers between 1 and t.
- All numbers in X are (distinct) integers between 1 and $t/2 - 1$.

Output

For each test case, print a line containing the minimum error $E(P)$.

Sample Input	Sample Output
2 10 3 4 1 7 4 2 3 4 20 4 5 1 3 4 9 8 1 5 9	2 0

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Problem J.
Mean Weight of a Critical Cycle

Time limit: 3 seconds

Intelligent Computer Programming Company (ICPC) is a well-established company for developing artificial intelligence, which attracts many young talents to join this strong team every year. ICPC has n divisions, denoted by $0, 1, \ldots, n-1$, and a work-flow for completing a task requested by a delegator. In the progress of executing a task, a division may need to plan and preprocess some subtasks, and then transfer this semi-finished task to another division. If Division i transfer a semi-finished task to another Division j, Division i is a predecessor of Division j. Moreover, let weight $w(i, j)$ be the cost for transferring a semi-finished task from Division i to Division j. We can represent a work-flow of ICPC using a directed graph $G = (V, E)$ with a weight function $w : E \mapsto \mathbb{Z}_+ \cup \{0\}$, in which n vertices represent n divisions, and a directed edge from Vertex i (*head*) to Vertex j (*tail*) exists if Division i is a predecessor of Division j. In addition, for two arbitrary vertices i and j in G, there is at most one directed edge between i and j.

Due to budget restriction, ICPC wants to improve the performance of the work-flow G by finding the minimum mean weight μ^* of all directed cycles in G. The mean weight of a cycle is defined as the summation of all the edge weights of the cycle divided by the number of edges. We call a cycle C for which the mean weight of C is equal to μ^* a *critical cycle*. Your task is to write a computer program to compute μ^* in an irreducible fraction form $\frac{p}{q}$ with p and q being non-negative integers, and output the numerator p and the denominator q; otherwise, if no critical cycle exists, please output -1.

For an example as shown in Figure 1, the bold cycle illustrated in Fig. 1(a) is a critical cycle C with $\mu^* = \frac{3+2+0}{3} = \frac{5}{3}$. The output is 5 and 3. On the other hand, the bold cycle illustrated in Fig. 1(b) is not a critical cycle.

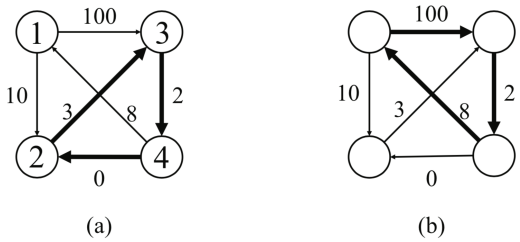


Figure 1: Examples of (a) a critical cycle and (b) a non-critical cycle.

Input

The first line of the input file contains an integer T that indicates the number of test cases as follows.

For each test case, the first line contains two integers (separated by a space) representing the number of divisions and the number of edges m. Then it is immediately followed by m lines, in which each line contains three integers that represent the head vertex, the tail vertex, and the corresponding weight; moreover, any two consecutive integers are separated by a space.

- $1 \leq T \leq 11$.
- $2 \leq n \leq 1000$ for each test case.
- The weight of each edge is non-negative and no larger than 1000000.

Output

The output contains one line for each test case. Each line contains two positive integers p and q, separated by a space, where p and q are the numerator and denominator of μ^*, respectively. Note that $\frac{p}{q}$ is an irreducible fraction. (i.e., the greatest common divisor of p and q is 1.)

Sample Input	Sample Output
3 4 6 1 2 10 1 3 100 2 3 3 3 4 2 4 1 8 4 2 0 10 10 1 2 1 2 3 1 3 4 1 4 5 1 5 6 1 6 7 1 7 8 1 8 9 1 9 10 1 10 1 1 2 1 1 2 1	5 3 1 1 -1

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Problem K.
Monochrome Land

Time limit: 1 second

Welcome to Monochrome Land! Here, chameleons would change their colors between black and white to hide themselves from predators. Their predators are two kinds of birds, namely the whites and the blacks, where the whites only eat white chameleons while the blacks only eat black chameleons. So, when a chameleon senses a bird of the same color is around, it will change its color immediately to the opposite one. Moreover, a chameleon will change its color when a neighboring chameleon of the same color does so. Mike, a naughty student, has prepared two fake birds, one for each color, where he would use them to turn all chameleons into the same color. What he would do is to show a fake bird to some chameleon, force it to change color, and wait until no more chameleon changes color. Then, he would repeat the above procedure again and again, until all chameleons are in the same color. Mike wants to know the minimum number of times he needs to show the fake bird. Can you help him?

Precisely, we can describe the problem as follows. The input is a simple undirected graph such that each node v represents a chameleon C_v, and each edge $\{u, v\}$ indicates that chameleons C_u and C_v corresponding to the endpoints are neighboring. Each node is colored black or white to represent the color of the corresponding chameleon at the moment. When a fake bird is shown to a chameleon, the corresponding node will change its color to the opposite one; moreover, for any node v, if a neighboring node of the same color changes color, v will change its color. A fake bird can be shown to an arbitrary node at a time, and force that node to switch color, which may then force some other neighboring nodes to switch color, and recursively force more nodes to switch color. Our target is to find out how many times a fake bird is shown so that all nodes are turned into the same color.

For simplicity, in this problem, the input graph is always connected.

Take Figure 1 as an example input. If Mike shows the white chameleon C_3 with a fake white bird, then chameleons C_3 and C_4 will switch their colors into black, leading to the situation shown in Figure 2. Now, if Mike further

shows any of the black chameleons (i.e., C_2, C_3, C_4 or C_5) a fake black bird, then all the chameleons will become white afterwards.

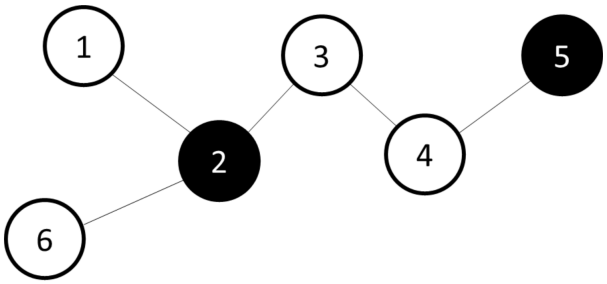


Figure 1: There are four white chameleons labeled as C_1, C_3, C_4, and C_6, while the black chameleons are labeled as C_2 and C_5 .

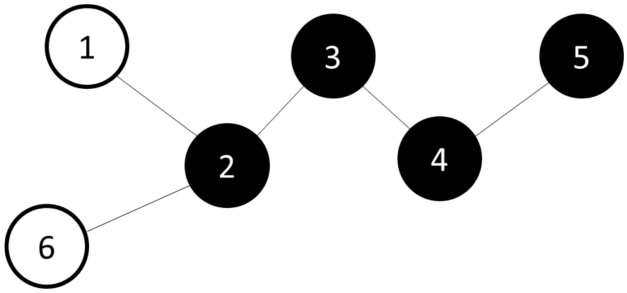


Figure 2: After showing chameleon C_3 in Figure 1 with a fake white bird.

Input

The first line of the input file contains an integer T indicating the number of test cases, each of them is described below. For each test case, The first line contains two space-separated integers N and E, which are the number of chameleons in the current graph and the number of neighboring relationship. The next line contains N characters separated by a space, which represent the color of chameleons in the order of $C_1, C_2, \ldots, C_N$. And then it is followed by E lines, each contains two integers u and v to represent the neighboring relationship between chameleons C_u and C_v.

- $1 \leq T \leq 10$
- $1 \leq N \leq 200$

Output

For each test case, please output the minimum times Mike needs to use a fake bird to turn all chameleons into the same color.

Sample Input	Sample Output
3 6 5 W B W W B W 1 2 2 3 2 6 3 4 4 5 6 6 W B W B W B 1 2 2 3 3 4 4 5 5 6 6 1 2 1 B B 1 2	2 3 0